



مختبر الإمارات للسلامة  
EMIRATES SAFETY LABORATORY

# TEST REPORT

REPORT NUMBER.  
0206-25-CR-01

## Reaction to Fire Classification Report of 4mm thick "Alubond USA FR A2" Aluminium Composite Panel

**According to:** EN 13501-1:2018

**Date of issue:** 02 October 2025



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# 1 Introduction

This classification report defines the classification assigned to 4mm thick “Alubond USA FR A2” Aluminium Composite Panel in accordance with the procedures given in EN 13501-1:2018.



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## CLASSIFICATION OF REACTION TO FIRE IN ACCORDANCE WITH EN 13501-1:2018

|                                  |                                                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Sponsor:</b>                  | Eurocon Building Industries FZE<br>Plot 2E-03, 2E-08, Hamriyah Freezone,<br>Sharjah, United Arab Emirates |
| <b>Prepared by:</b>              | Emirates Safety Laboratory<br>Warsan Third, Dubai, United Arab Emirates                                   |
| <b>Notified Body No:</b>         | Not applicable                                                                                            |
| <b>Test performed at:</b>        | Emirates Safety Laboratory<br>Warsan Third, Dubai, United Arab Emirates                                   |
| <b>Product name:</b>             | 4mm thick “Alubond USA FR A2” Aluminium Composite Panel                                                   |
| <b>Classification report no.</b> | 0206-25-CR-01                                                                                             |
| <b>Issue number:</b>             | 1                                                                                                         |
| <b>Date of issue:</b>            | 02 October 2025                                                                                           |

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# 2 Details of the Classified Product

## 2.1 General

The product is defined by the test sponsor as Aluminium Composite Panel.

## 2.2 Product Description

The product, 4mm thick “Alubond USA FR A2” Aluminium Composite Panel is described in support of the classification listed in Section 4.

## Note:

1: The test sponsor does not want to disclose the information; thus, this was kept confidential by ESL.

<sup>(A)</sup>: The information is as declared by the client and ESL Certification; ESL Testing was unable to verify the information.

<sup>(B)</sup>: The information is as measured by ESL Testing.

|                     |                                                                                                                         |                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| General description | Aluminium Composite Panel (ACP)                                                                                         |                                                                                           |
| Product name        | "Alubond USA FR A2" <sup>(A)</sup>                                                                                      |                                                                                           |
| Manufacturer        | Eurocon Building Industries FZE<br>Plot 2E-03, 2E-08, Hamriyah Freezone,<br>Sharjah, United Arab Emirates               |                                                                                           |
| Thickness           | 4mm <sup>(A)</sup> ; 4.19mm <sup>(B)</sup>                                                                              |                                                                                           |
| Area weight         | 8.2 ± 0.5 kg/m <sup>2(A)</sup> ;<br>8.8 kg/m <sup>2(B)</sup> (measured by the laboratory from a 250mm x 250mm specimen) |                                                                                           |
| Topcoat             | Type                                                                                                                    | Polyvinylidene fluoride (PVDF) <sup>(A)</sup>                                             |
|                     | Manufacturer                                                                                                            | Spectrum Industries (L.L.C) <sup>(A)</sup>                                                |
|                     | Worst colour tested                                                                                                     | Black – as chosen by ESL Certification from Table 2 of EN ISO 1716 report (0206-25-TR-01) |
|                     | Thickness                                                                                                               | 18-20microns <sup>(A)</sup>                                                               |
|                     | Density                                                                                                                 | 1700kg/m <sup>3(A)</sup>                                                                  |
| Primer              | Type                                                                                                                    | Polyvinylidene fluoride (PVDF) <sup>(A)</sup>                                             |
|                     | Manufacturer                                                                                                            | Spectrum Industries (L.L.C) <sup>(A)</sup>                                                |
|                     | Thickness                                                                                                               | 6-7microns <sup>(A)</sup>                                                                 |
|                     | Density                                                                                                                 | 1170 kg/m <sup>3(A)</sup>                                                                 |
| Aluminium sheet     | Description                                                                                                             | Mill finish aluminium coil <sup>(A)</sup>                                                 |
|                     | Thickness                                                                                                               | 0.5 ± 0.05mm <sup>(A)</sup>                                                               |
|                     | Density                                                                                                                 | 2720 kg/m <sup>3(A)</sup>                                                                 |
| Adhesive            | Description                                                                                                             | Polymer based adhesive film <sup>(A)</sup>                                                |
|                     | Manufacturer                                                                                                            | Alubond Middle East FZE                                                                   |
|                     | Thickness                                                                                                               | 80microns <sup>(A)</sup>                                                                  |
|                     | Density                                                                                                                 | 938 ± 20 kg/m <sup>3(A)</sup>                                                             |
| Core                | Description                                                                                                             | Mineral core <sup>(A)</sup>                                                               |
|                     | Manufacturer                                                                                                            | Metalplast Industries FZE <sup>(A)</sup>                                                  |
|                     | Thickness                                                                                                               | 3.0 ± 0.1mm <sup>(A)</sup> ; ; 3.10mm <sup>(B)</sup>                                      |
|                     | Density                                                                                                                 | 1850 ± 100 kg/m <sup>3(A)</sup> ; 1845 kg/m <sup>3(B)</sup>                               |
| Adhesive            | Description                                                                                                             | Polymer based adhesive film <sup>(A)</sup>                                                |
|                     | Manufacturer                                                                                                            | Alubond Middle East FZE                                                                   |
|                     | Thickness                                                                                                               | 80microns <sup>(A)</sup>                                                                  |
|                     | Density                                                                                                                 | 938 ± 20 kg/m <sup>3(A)</sup>                                                             |
| Aluminium sheet     | Description                                                                                                             | Mill finish aluminium coil <sup>(A)</sup>                                                 |
|                     | Thickness                                                                                                               | 0.5 ± 0.05mm <sup>(A)</sup>                                                               |
|                     | Density                                                                                                                 | 2720 kg/m <sup>3(A)</sup>                                                                 |
| Service coat        | Type                                                                                                                    | Polyester (PE) Monocoat <sup>(A)</sup>                                                    |
|                     | Manufacturer                                                                                                            | Spectrum Industries (L.L.C) <sup>(A)</sup>                                                |

|                                                                                                                                                                                                        |                             |                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------|
|                                                                                                                                                                                                        | Thickness                   | 8-10 microns <sup>(A)</sup>                                  |
|                                                                                                                                                                                                        | Density                     | 1100 kg/m <sup>3(A)</sup>                                    |
| Dimensions of the test specimens as measured by the laboratory, see Section 4.3<br>(Each specimen consisted of two wings mounted into an aperture in a specimen trolley forming a vertical 90° corner) | Long wing (width x height)  |                                                              |
|                                                                                                                                                                                                        | Panel A                     | 792.5 x 992.5 mm                                             |
|                                                                                                                                                                                                        | Panel B                     | 196.5 x 992.5 mm                                             |
|                                                                                                                                                                                                        | Panel C                     | 792.5 x 492.5 mm                                             |
|                                                                                                                                                                                                        | Panel D                     | 196.5 x 492.5 mm                                             |
|                                                                                                                                                                                                        | Short wing                  | 495 x 1500mm (width x height)                                |
| Backing board                                                                                                                                                                                          | Material and classification | Calcium Silicate board<br>A2-s1, d0 based on EN 13501-1:2018 |
|                                                                                                                                                                                                        | Thickness, mm               | 12                                                           |
|                                                                                                                                                                                                        | Density, kg/m <sup>3</sup>  | 870                                                          |

## 2.3 Reports

| Name of laboratory         | Name of sponsor                 | Report ref. no.                  | Test method and date  |
|----------------------------|---------------------------------|----------------------------------|-----------------------|
| Emirates Safety Laboratory | Eurocon Building Industries FZE | 0206-25-TR-01<br>02 October 2025 | EN ISO 1716:2018      |
|                            |                                 | 0206-25-TR-02<br>02 October 2025 | EN 13823:2020+A1:2022 |

## 2.4 Results

| Test method and test number         | Parameter                                                                                                    | No. of tests | TEST RESULTS                |                            |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------|-----------------------------|----------------------------|
|                                     |                                                                                                              |              | Continuous parameter - mean | Compliance with parameters |
| EN ISO-1716:2018<br>(0206-25-TR-01) | External non-substantial layer<br>(Topcoat + Primer)<br>$Q_{PCS} \text{ MJ/m}^2 \leq 4.0$<br>$\text{MJ/m}^2$ | 3            | 0.7                         | Compliant                  |
|                                     | External non-substantial layer<br>(Back coat)<br>$Q_{PCS} \text{ MJ/m}^2 \leq 4.0$<br>$\text{MJ/m}^2$        |              | 0.2                         |                            |
|                                     | Substantial Layer<br>(Core)                                                                                  |              | 1.7                         |                            |

|                                             |                                                                                                                             |   |                    |           |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---|--------------------|-----------|
|                                             | $Q_{\text{pcs}} \text{ MJ/kg} \leq 3.0$<br>MJ/kg                                                                            |   |                    |           |
|                                             | <i>Substantial Layer</i><br><i>(Aluminium)</i><br>$Q_{\text{pcs}} \text{ MJ/kg} \leq 3.0$<br>MJ/kg                          |   | 0                  |           |
|                                             | <i>Internal non-substantial layer</i><br><i>(Adhesive)</i><br>$Q_{\text{pcs}} \text{ MJ/m}^2 \leq 4.0$<br>MJ/m <sup>2</sup> |   | 3.5                |           |
|                                             | <i>Product as a whole</i><br>$Q_{\text{pcs}} \text{ MJ/kg} \leq 3.0$<br>MJ/kg                                               |   | 2.0                |           |
|                                             |                                                                                                                             |   |                    |           |
| EN<br>13823:2020+A1:2022<br>(0206-25-TR-02) | <b>For Class “A2”</b>                                                                                                       |   |                    |           |
|                                             | $\text{THR}_{600\text{s}} \leq 7.5 \text{ MJ}$                                                                              | 3 | 0.7                | Compliant |
|                                             | $\text{FIGRA}_{0.2\text{MJ}} \leq 120 \text{ W/s}$                                                                          | 3 | 1                  | Compliant |
|                                             | Lateral flame spread<br>< Edge of specimen                                                                                  | 3 | < Edge of specimen | Compliant |
|                                             | <b>For subclass “s1”</b>                                                                                                    |   |                    |           |
|                                             | $\text{TSP}_{600\text{s}} \leq 50 \text{ m}^{2(\text{B})}$                                                                  | 3 | 11                 | Compliant |
|                                             | $\text{SMOGRA} \leq 30 \text{ m}^2/\text{s}^{2(\text{B})}$                                                                  | 3 | 0                  | Compliant |
|                                             | <b>For subclass “d0”</b>                                                                                                    |   |                    |           |
|                                             | No flaming droplets/ particles within 600s                                                                                  | 3 | Not observed       | Compliant |

### 3 Classification and Field of Application

#### 3.1 Reference of classification

This classification has been carried out in accordance with EN 13501-1:2018.

#### 3.2 Classification

The product, 4mm thick "Alubond USA FR A2" Aluminium Composite Panel, in relation to its reaction to fire behaviour is classified as:

A2

The additional classification in relation to smoke production is:

s1

The additional classification in relation to flaming droplets/particles is:

d0

The format of the reaction to fire classification for construction products excluding floorings and linear pipe thermal insulation products is:

| Fire behaviour |   | Smoke production |   | Flaming droplets |
|----------------|---|------------------|---|------------------|
| A2             | - | s                | 1 | , d 0            |

Reaction to fire classification: **A2-s1, d0**

### 3.3 Field of application

The classification is valid for the following product parameters:

|                                              |                                                                             |
|----------------------------------------------|-----------------------------------------------------------------------------|
| Colour of topcoat                            | Any variation in the tested type                                            |
| Type of topcoat and primer                   | No variation allowed                                                        |
| Thickness and area density of the components | No variation allowed                                                        |
| Thickness and area density of the product    | No variation allowed                                                        |
| Product composition and layout               | No variation allowed                                                        |
| Joints                                       | For assembly without joints, and with joints up to the tested configuration |
| Air gap                                      | Ventilated                                                                  |

This classification is valid for the following end-use applications:  
Details of buildup/Construction - No variation in the build-up allowed.

### 3.4 Limitations

This report does not represent type approval or certification of the product.

## 4 Signatories

| Prepared by                                                                         | Reviewed by                                                                         | Authorised by                              |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Rachel Marie Novelo</b><br>Reaction to Fire<br>Principal Engineer                | <b>Sebastian Ukleja</b><br>Testing Manager                                          | <b>Sebastian Ukleja</b><br>Testing Manager |
| Signature                                                                           | Signature                                                                           | Signature                                  |
|  |  |                                            |

—END OF REPORT—